NATIONAL INSTITUTE OF TECHNOLOGY, ARUNACHALPRADESH
DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

MINOR PROJECT 2

Design and Implementation of an All-Optical 2×1 Multiplexer Using Cross-Phase Modulation in Highly Nonlinear Fiber

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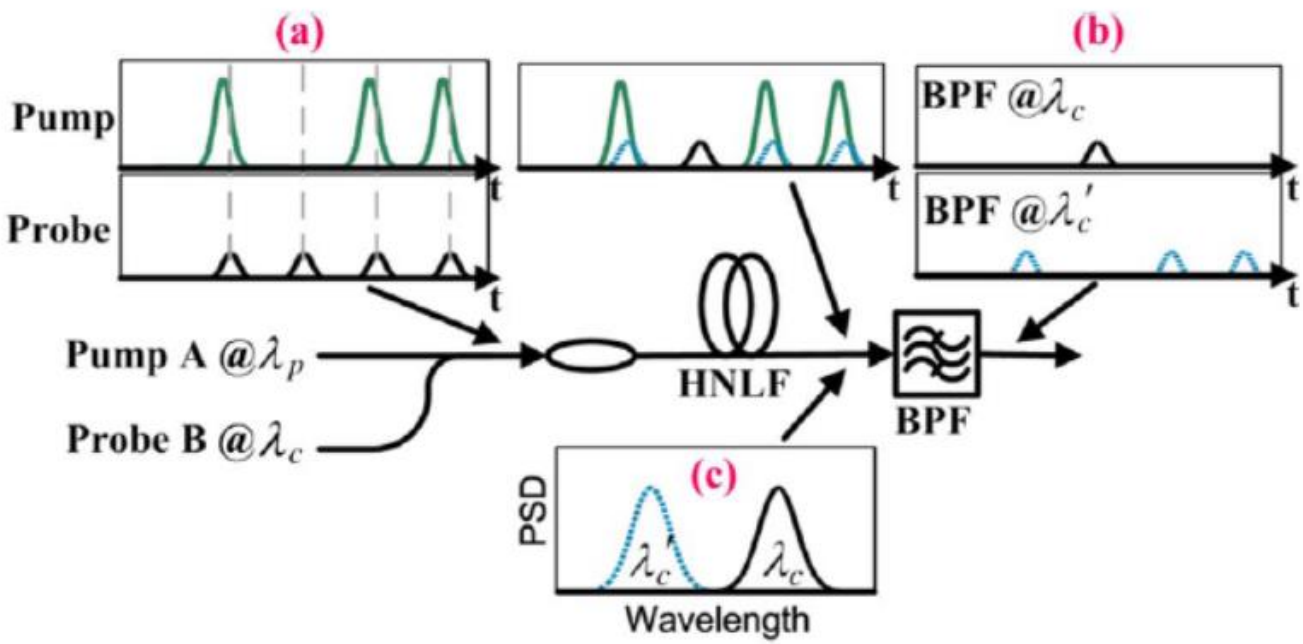
Working Principle Based on Cross-Phase Modulation (XPM):

- When the high-power Signal(A) **(Pump)** and low-power signal(B) **(Probe)** pass through HNLF(Highly Non Linear fiber), the pump's intensity imposes a phase shift on the probe through XPM under HNLF, resulting in a wavelength shift of Probe.
- Using Band-Pass Filters (BPFs), the shifted or unshifted probe components are separated to generate logical outputs.
- (a) Pump and probe pulses enter HNLF.
- (b) XPM causes the probe to shift only when pump = 1 → filtered as logic output.
- (c) Spectrum shows original λc and XPM-shifted λ'c.



Cross-Phase Modulation (XPM)

Operation: Logic Table Representation

Pump A	Probe B	XPM effect	Output Wavelength
0	0	No XPM	No Probe
0	1	No XPM	λc (original)
1	0	No probe → no shift	No Probe
1	1	Probe gets phase-shift → wavelength shift	λ'c (shifted)

Mathematical Understanding of Cross Phase Modulation and Calculation of Shifted Wavelength

This is the standard XPM formula for calculation of Wavelength Shift (Δλ)

$$\Delta\lambda \approx \frac{\lambda^2}{c}\left(\frac{2\gamma L_{\text{eff}} P_{\text{peak}}}{T_{\text{rise}}}\right)$$

These are values we have taken in according to our fiber

Length of fiber= 1km
α = 0.2 dB/km(Attenuation) → Effective Length ≈ **0.98 km**
γ ≈ **9.6 /W/km** (nonlinearity factor)
T Rise = 3.1 ps (typical in NRZ/RZ bitrates)
Pump peak power = **10 dBm = 0.01 W**

Shifted Wavelength Results after applying the formula

λ_A=1555.2 nm ⟶ Shifted λ_A'=1555.2 ± 3 nm
λ_B=1541.3 nm ⟶ Shifted λ_B'=1541.3 ± 3 nm

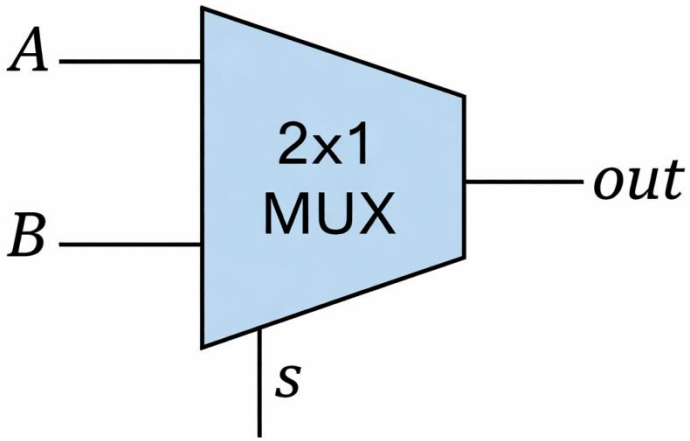
Red Shift λ= λ+Δλ Blue Shift λ= λ-Δλ

Reason behind Red and Blue shift

The pump power changes over time (rising edge + falling edge).
During the rising edge, wavelength shifts one way (red).
During the falling edge, wavelength shifts the opposite way (blue).

Aim: Implementing of All-Optical 2X1 Multiplexer

OUT = S'·A + S·B



S	OUT
0	A
1	B

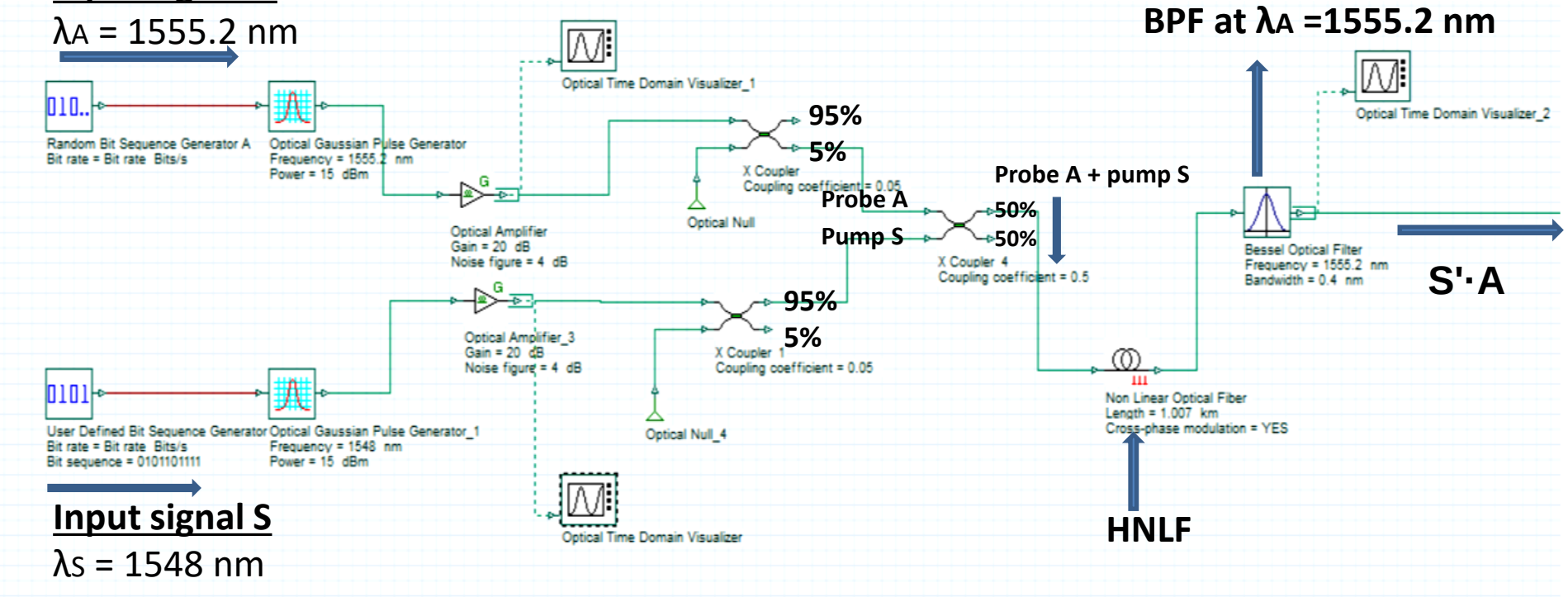
Steps of Implementation

1. Extraction of Logic **S'·A**
2. Extraction of Logic **S·B**
3. Addition of the both

Experimental Setup For Extraction of Logic **S'·A**

Input signal A

λ_A = 1555.2 nm



Input signal S

λ_S = 1548 nm

95% = Pump(High Power)

5% = Probe(Low Power)

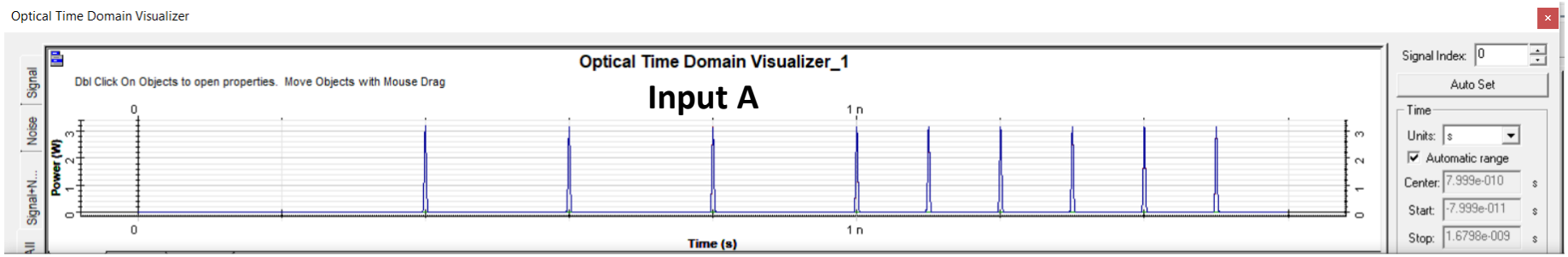
Optisystem Layout

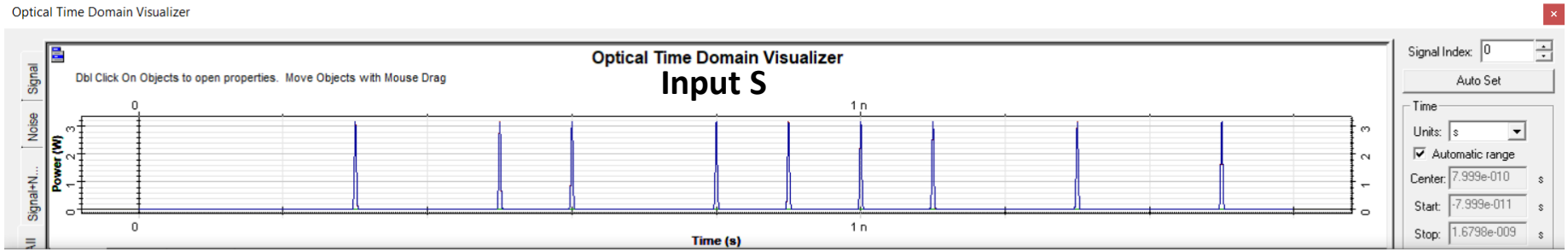
Logic Table Output (BPF sees A as probe and S as pump)

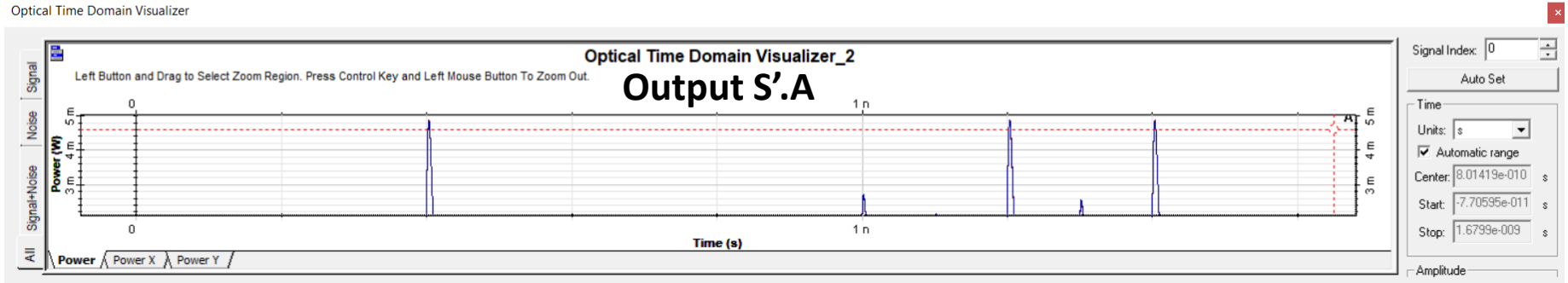
To extract the logic S'.A must pass the wavelength Corresponding to input S=0,A=1 i.e. we need pass **λ_A(Original Probe)**

Input S (Pump)	Input A (Probe)	Spectral component at BPF input	BPF tuned to λ_A = 1555.2 nm (passes original probe)→output
0	0	No Probe	0
0	**1**	**λ_A(Original Probe)**	**1**
1	0	No Probe	0
1	1	λ_A'(Shifted Probe)	0

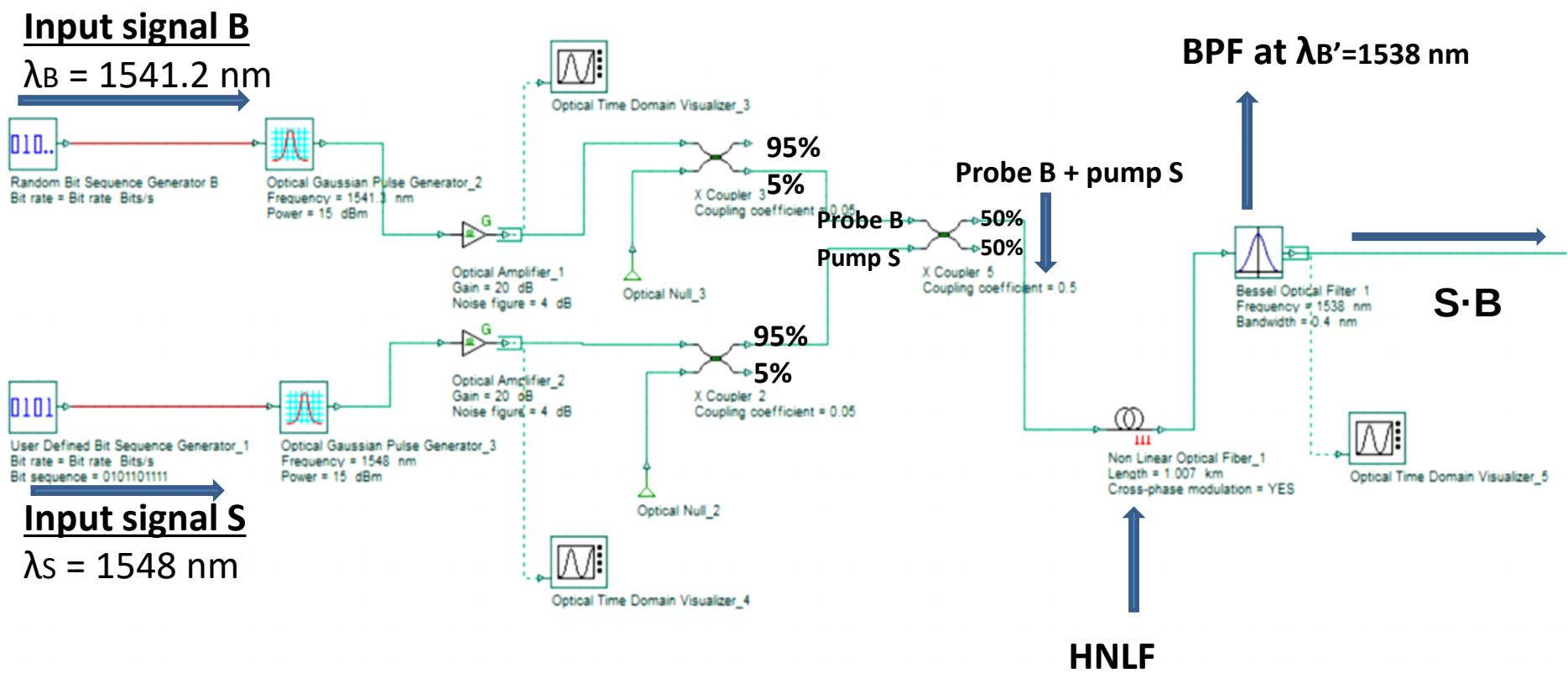
Experimental Result of Extraction of Logic **S'·A**







Experimental Setup For Extraction of Logic **S·B**



95% = Pump(High Power)
5% = Probe(Low Power)

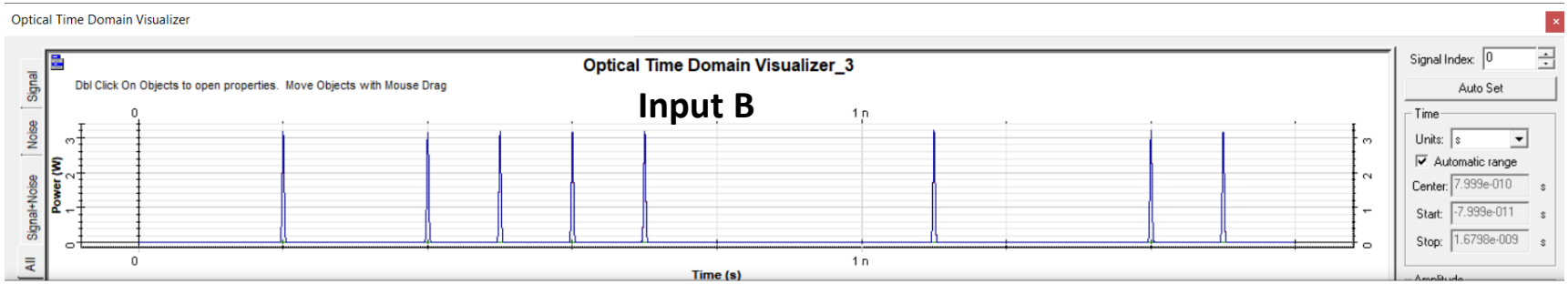
Optisystem Layout

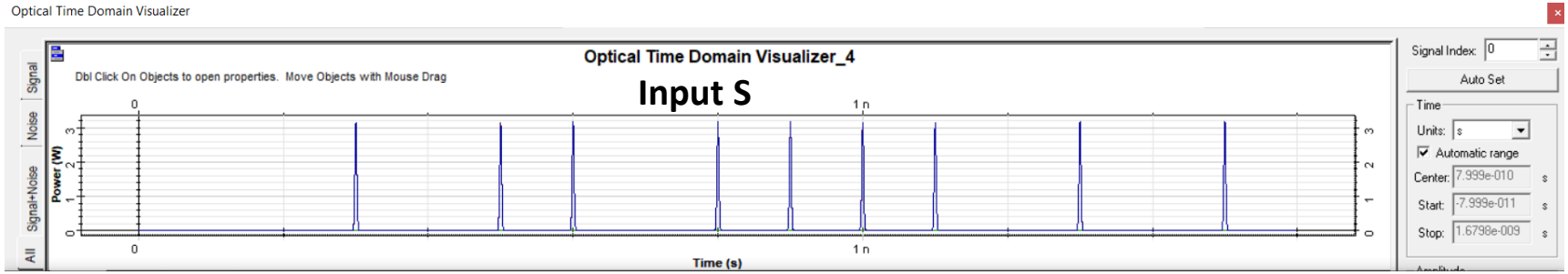
Logic Table Output (BPF sees B as probe and S as pump)

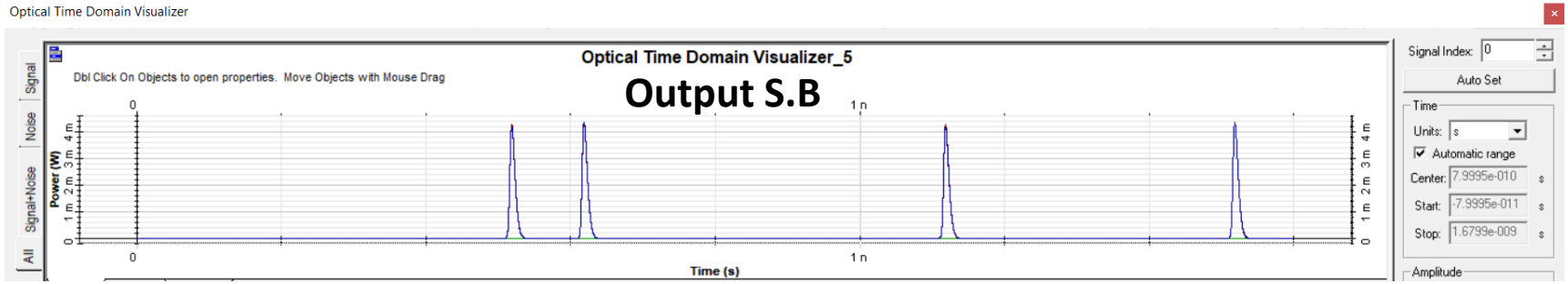
To extract the logic S.B must pass the wavelength Corresponding to input S=1,B=1 i.e. we need pass **λ_B'(Shifted Probe)**

Input S (Pump)	**Input B (Probe)**	**Spectral component at BPF input**	**BPF tuned to λ_B' = 1538 nm (passes shifted probe)→output**
0	0	No Probe	0
0	1	λ_B(Original Probe)	0
1	0	No Probe	0
1	**1**	**λ_B'(Shifted Probe)**	**1**

Experimental Result of Extraction of Logic **S·B**



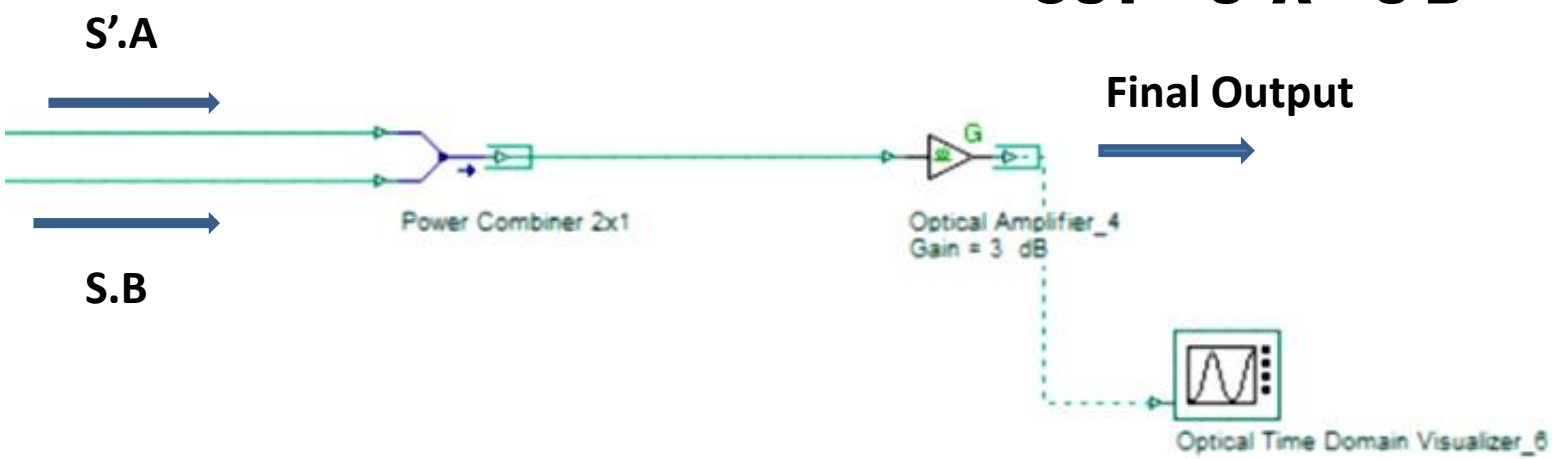




Addition of logic S’.A and S.B

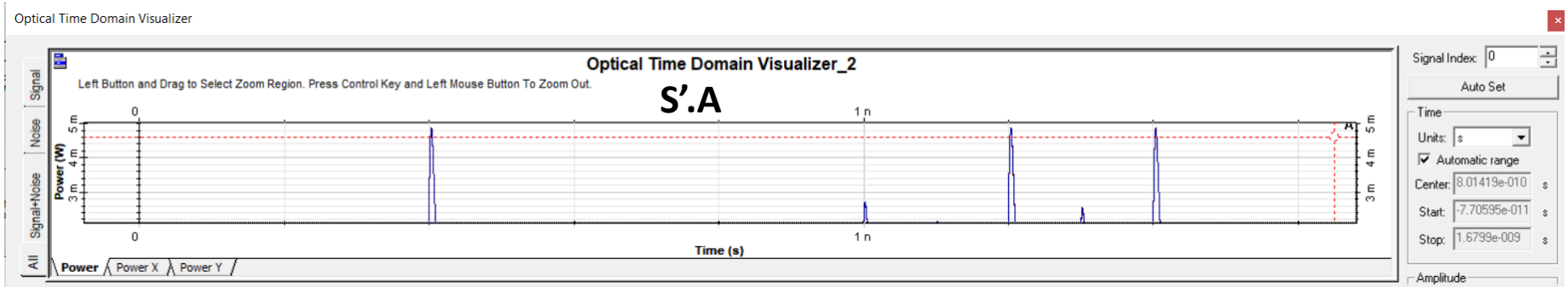
Since only one of the signals (either A or B) appears at the output at any given time, a power combiner can be used directly instead of an OR-based HNLF model

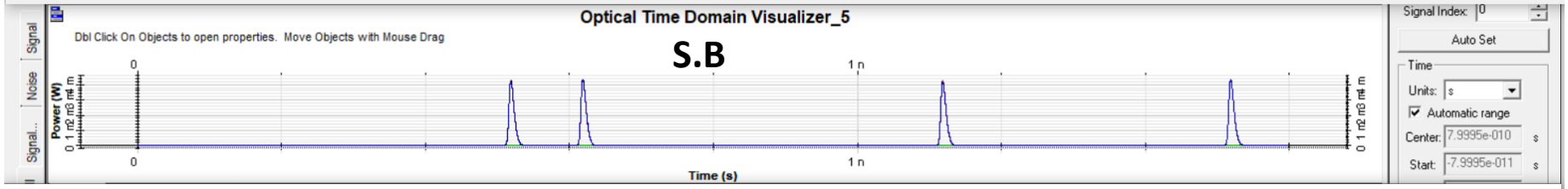
OUT = S'·A + S·B

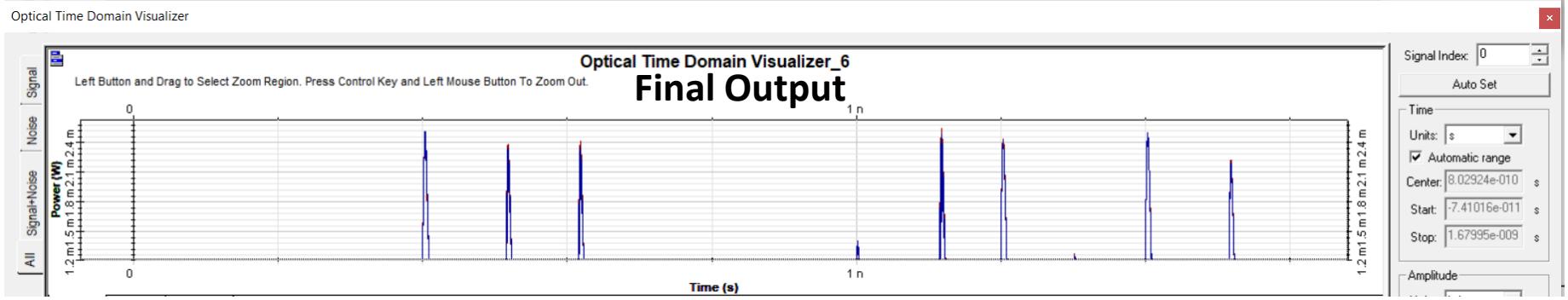


Optisystem Layout

Experimental Result of Final Output Signal







Application in Reversible Computing

What is Reversible computing ?

A system in which it is possible to reconstruct the original input from the output.

As a result, reversible computing avoids information loss and reduces energy dissipation.

In Optical computing, logic values are represented using different wavelengths (λ).Since each wavelength is unique and distinguishable, we can trace back the output to the exact input combination.

Input A	Input B	Output Spectral Component
0	0	No Signal
0	1	λ_B (Original Signal B)
1	0	λ_A (Original Signal A)
1	1	$\lambda_{B'}$ / $\lambda_{A'}$ (Shifted Signals)

Future Works :

Extended Logic Design Using Optical Techniques

Using the same optical technique, more complex **combinational logic circuits** can be designed efficiently.

Design of Sequential Logic and Optical Memory

By introducing **optical time delay** and incorporating a **feedback loop from output to input**, it is possible to design **sequential logic circuits and optical memory systems**.

Reference:

Jifang Qiu, Kai Sun, Martin Rochette, Lawrence R. Chen,
"Reconfigurable All-Optical Multilogic Gate (XOR, AND, OR) Based on XPM in HNLF,"
IEEE Photonics Technology Letters, 2010.